

INTRODUCTION AND OVERVIEW

Consider “nice” subsets of Euclidean spaces, defined by conditions like

$$(*) \quad f_1(x) = f_2(x) = \cdots = f_k(x) = 0, \quad g_1(x) > 0, \dots, g_l(x) > 0,$$

$x = (x_1, \dots, x_n) \in \mathbb{R}^n$, for “nice” functions $f_1, \dots, f_k, g_1, \dots, g_l$. Now perform elementary logical, geometric and topological operations on these sets: take unions, intersections, complements, closures and cartesian products, and project into lower-dimensional euclidean spaces. If we keep repeating these operations with the new sets that arise, then, roughly speaking, two kinds of things can happen:

- (1) After only a few such operations, say after taking projections of finite unions of the sets one starts out with, a *stabilization* occurs, and performing further operations does not produce any new sets.
- (2) Ever more complicated sets arise: Cantor-like sets, Borel sets of arbitrarily high complexity, and so on. (For an example, see Chapter 1, (2.6).)

It is phenomenon (1) that is of particular interest to us: here we remain in the realm of geometry and topology envisaged by Poincaré, and we may view the study of the corresponding category of “nice” spaces and maps between them as an unfolding of the rich algebraic-analytic-topological structure of the continuum.

When phenomenon (2), on the other hand, is pursued, then the ties to geometry tend to become very loose. We enter here *Cantor’s paradise*: only set-theoretic features ultimately survive in this bleak landscape. Already on the Borel level the real line and the real plane (or Cantor space for that matter) become indistinguishable.

Despite these large contrasts it is often hard to determine for a specific class of “nice functions” whether we are in case (1) or in case (2). In Chapter 1 we show that with linear functions we are in case (1), and in Chapter 2 we do the same for polynomial functions. But only in a later volume shall we pay special attention to the art of proving that many interesting situations fall under phenomenon (1). **The emphasis in this book is on developing the general theory of what else is true if one happens to be in case (1).**

A key example of phenomenon (1) is provided by the semialgebraic sets. A **semialgebraic** set in $\mathbb{R}^n$ is a finite union of sets of the form $(*)$ where $f_1, \dots, f_k, g_1, \dots, g_l \in \mathbb{R}[X]$, $X = (X_1, \dots, X_n)$. (If the polynomials f_i and g_j are also assumed to be of degree ≤ 1 we speak of **semilinear** sets.) In particular, for $n = 1$, we get just the finite unions of intervals and points, or equivalently, the subsets of the real line with only finitely many connected components.

A major fact about semialgebraic sets is the Tarski-Seidenberg projection property:

If $A \subseteq \mathbb{R}^{n+1}$ is semialgebraic, then $\pi(A) \subseteq \mathbb{R}^n$ is semialgebraic, where $\pi: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$ is the projection map on the first n coordinates.

Related results are: the closure, interior, and convex hull of a semialgebraic set are semialgebraic; a semialgebraic set has only finitely many connected components, each of which is also semialgebraic; a semialgebraic set can be triangulated, and stratified into finitely many semialgebraic real analytic manifolds. (All of this remains true with “semilinear” instead of “semialgebraic”.) A proper setting for many results of this kind is provided by the theory of o-minimal structures, as this book will attempt to show.

O-minimal structures on the real line

The class of semialgebraic sets and the class of semilinear sets, are both examples of an o-minimal structure on $\mathbb{R}$, the ordered set of real numbers (the real line).

DEFINITION. An **o-minimal structure on $\mathbb{R}$** is a sequence $\mathcal{S} = (\mathcal{S}_n)_{n \in \mathbb{N}}$ such that for each n :

- (1) $\mathcal{S}_n$ is a boolean algebra of subsets of $\mathbb{R}^n$, that is, $\mathcal{S}_n$ is a collection of subsets of $\mathbb{R}^n$, $\emptyset \in \mathcal{S}_n$ and if $A, B \in \mathcal{S}_n$, then $A \cup B \in \mathcal{S}_n$, and $\mathbb{R}^n - A \in \mathcal{S}_n$;
- (2) $A \in \mathcal{S}_n \Rightarrow A \times \mathbb{R} \in \mathcal{S}_{n+1}$ and $\mathbb{R} \times A \in \mathcal{S}_{n+1}$;
- (3) $\{(x_1, \dots, x_n) \in \mathbb{R}^n : x_i = x_j\} \in \mathcal{S}_n$ for $1 \leq i < j \leq n$;
- (4) $A \in \mathcal{S}_{n+1} \Rightarrow \pi(A) \in \mathcal{S}_n$, where $\pi: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$ is the usual projection map;
- (5) $\{r\} \in \mathcal{S}_1$ for each $r \in \mathbb{R}$, and $\{(x, y) \in \mathbb{R}^2 : x < y\} \in \mathcal{S}_2$;
- (6) the only sets in $\mathcal{S}_1$ are the finite unions of intervals and points. (“Interval” always means “open interval”, with infinite endpoints allowed.)

The first four axioms guarantee that each set $A \subseteq \mathbb{R}^n$ that is definable (in a precise logical sense) from the sets in $\mathcal{S}$ also belongs to $\mathcal{S}$. (See Chapter 1.) With axiom (5) they imply that finite unions of intervals and points belong to $\mathcal{S}_1$. Axiom (6) says that no further subsets of $\mathbb{R}$ belong to $\mathcal{S}_1$; this is the minimality axiom that explains the term “o-minimal”, “o” standing for “order”. This o-minimality axiom can be viewed as expressing compatibility with the order (and hence topology) of the real line, and is clearly the simplest compatibility condition of this nature.

Examples of o-minimal structures on the real line

- the semilinear sets;
- the semialgebraic sets;
- the subsets of the affine spaces $\mathbb{R}^n$ for $n = 0, 1, 2, \dots$ that are subanalytic in the larger projective space $\mathbb{P}^n(\mathbb{R})$;

- the images in $\mathbb{R}^n$ for $n = 0, 1, 2, \dots$ under projection maps $\mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ of sets of the form $\{(x, y) \in \mathbb{R}^{n+k} : P(x, y, e^x, e^y) = 0\}$ where P is a real polynomial in $2(n+k)$ variables, and where $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_k)$, $e^x = (e^{x_1}, \dots, e^{x_n})$, $e^y = (e^{y_1}, \dots, e^{y_k})$.

These examples and others will be treated in some detail in this book and its planned sequel: the first two in Chapters 1 and 2 of this book, the others in a later volume. The o-minimality of the last example follows from a remarkable theorem proved by Wilkie in 1991. In contrast to the first three examples it contains the graph of a function on $\mathbb{R}$ growing faster at ∞ than any polynomial, namely e^x . There are many other interesting o-minimal structures on $\mathbb{R}$, but not too many. For instance, a result of Peterzil [45] says that there is exactly one o-minimal structure on $\mathbb{R}$ that lies strictly between the class of semilinear sets and the class of semialgebraic sets; this intermediate structure contains every bounded semialgebraic set, but not the graph of multiplication.

General facts about arbitrary o-minimal structures on the real line

From now on in this introduction we fix an o-minimal structure $\mathcal{S}$ on $\mathbb{R}$.

Let $A \subseteq \mathbb{R}^m$ and $f: A \rightarrow \mathbb{R}^n$. We say A is **definable** if $A \in \mathcal{S}_m$. The map f is said to be **definable** if its graph $\Gamma(f) \subseteq \mathbb{R}^{m+n}$ is definable. If f is definable, then the domain A of f and its image $f(A) \subseteq \mathbb{R}^n$ are also definable. The closure $\text{cl}(A)$ and interior $\text{int}(A)$ of a definable set $A \subseteq \mathbb{R}^n$ are definable. There are many more such elementary but useful facts, and Chapter 1 is basically about them. A less easy but fundamental result, proved in Chapter 3, is the following:

MONOTONICITY THEOREM.

If the function $f: (a, b) \rightarrow \mathbb{R}$ is definable, then there are $a = a_0 < a_1 < \dots < a_N = b$ such that f is continuous on each subinterval (a_i, a_{i+1}) , and either constant, or strictly increasing, or strictly decreasing, on each subinterval (a_i, a_{i+1}) .

This result is used throughout the subject. In particular it plays a key role in the proof that definable sets can be partitioned into finitely many *cells*. Cells are nonempty definable sets of an especially simple form. They are defined inductively as follows:

- (i) the cells in $\mathbb{R}^1 = \mathbb{R}$ are just the points $\{r\}$, and the intervals (a, b) ;
- (ii) let $C \subseteq \mathbb{R}^n$ be a cell; if $f, g: C \rightarrow \mathbb{R}$ are definable continuous functions such that $f < g$ on C , then

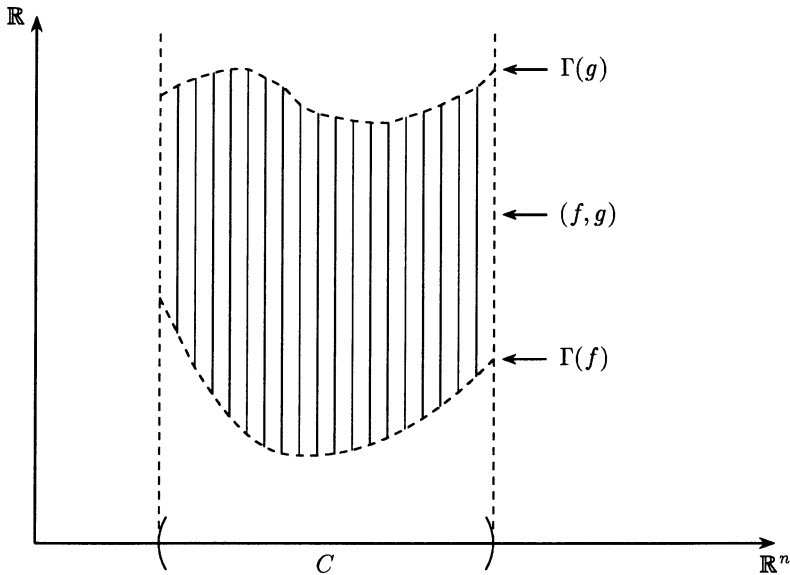
$$(f, g) := \{(x, r) \in C \times \mathbb{R} : f(x) < r < g(x)\}$$

is a cell in $\mathbb{R}^{n+1}$; also, given a definable continuous function $f: C \rightarrow \mathbb{R}$ the

sets

$$\begin{aligned}\Gamma(f), \\ (-\infty, f) &:= \{(x, r) \in C \times \mathbb{R} : r < f(x)\}, \\ (f, +\infty) &:= \{(x, r) \in C \times \mathbb{R} : f(x) < r\},\end{aligned}$$

are cells in $\mathbb{R}^{n+1}$; finally $C \times \mathbb{R} \subseteq \mathbb{R}^{n+1}$ is a cell. (See figure.)



It is extremely useful in inductive proofs to view the cells defined in (ii) as fiber spaces over their projection C , and in general to view $\mathbb{R}^{n+1}$ as fibered over $\mathbb{R}^n$ via the projection map $\pi : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$ on the first n coordinates. The next result is also established in Chapter 3 and is central in the subject. It is to some extent the many-variable version of the Monotonicity Theorem.

CELL DECOMPOSITION THEOREM.

Each definable set $A \subseteq \mathbb{R}^m$ has a finite partition $A = C_1 \cup \dots \cup C_k$ into cells C_i . If $f : A \rightarrow \mathbb{R}^n$ is a definable map, this partition of A can moreover be chosen such that all restrictions $f|_{C_i}$ are continuous.

It follows in particular that a definable set has only finitely many connected components, and these components are again definable (and path connected).

Definable invariants (Chapter 4)

The **dimension** of a cell is defined inductively in an obvious way, and then extended to all definable sets $A \subseteq \mathbb{R}^m$ by $\dim A := \max\{\dim(C) : C \subseteq A \text{ is a cell}\}$ if A is nonempty, while $\dim(\emptyset) = -\infty$. This dimension has good properties:

- (1) Let $f: A \rightarrow \mathbb{R}^n$ be a definable map, not necessarily continuous. Then $\dim A \geq \dim f(A)$; in particular $\dim A = \dim f(A)$ if f is injective.
- (2) $\dim(\text{cl}(A) - A) < \dim A$ for nonempty definable $A \subseteq \mathbb{R}^m$.

A more subtle “definable invariant” is the **Euler characteristic**. For a cell C of dimension d it takes the value $E(C) := (-1)^d$, and for an arbitrary definable set A we put

$$E(A) := \sum E(C_i)$$

where $C_1, \dots, C_k$ are the members of a finite partition of A into cells. One proves easily that this definition makes sense. More difficult is the following:

If $f: A \rightarrow \mathbb{R}^n$ is an injective definable map (not necessarily continuous!), then

$$E(A) = E(f(A)).$$

Results on definable families and definable collections (Chapters 4, 5)

Let $S \subseteq \mathbb{R}^{m+n}$ be a definable set. Then S gives rise to a family $(S_x)_{x \in \mathbb{R}^m}$ of definable subsets of $\mathbb{R}^n$, where $S_x := \{y \in \mathbb{R}^n : (x, y) \in S\}$. We call this a **definable family** (with parameter space $\mathbb{R}^m$), and we are interested in how the properties of S_x depend on x . Cell decomposition gives a uniform bound on the number of cells into which S_x can be partitioned, independent of x , and a more precise analysis along these lines leads to:

- (1) For each $d \in \{0, \dots, n\}$ the set $S(d) := \{x \in \mathbb{R}^m : \dim(S_x) = d\}$ is definable and

$$\dim\{(x, y) \in S : \dim(S_x) = d\} = \dim(S(d)) + d;$$

- (2) $E(S_x)$ takes only finitely many values as x runs through $\mathbb{R}^m$. Moreover, for each integer e the set $A(e) := \{x \in \mathbb{R}^m : E(S_x) = e\}$ is definable, and

$$E\{(x, y) \in S : x \in A(e)\} = e \cdot E(A(e)).$$

The collection of sets $\mathcal{C} := \{S_x : x \in \mathbb{R}^m\}$ is called a **definable collection**. (We distinguish it from the **family** $(S_x)_{x \in \mathbb{R}^m}$.) This collection has a surprising combinatorial property, as we explain now.

For each finite set $F \subseteq \mathbb{R}^n$, put $F \cap \mathcal{C} := \{F \cap S_x : x \in \mathbb{R}^m\}$, so $0 \leq |F \cap \mathcal{C}| \leq 2^{|F|}$. Now define a growth function $g_S: \mathbb{N} \rightarrow \mathbb{N}$ by

$$g_S(k) := \max\{|F \cap \mathcal{C}| : F \text{ is a } k\text{-element subset of } \mathbb{R}^n\},$$

so that $0 \leq g_S(k) \leq 2^k$. A remarkable fact is:

(3) *The function $g_S(k)$ is bounded by a polynomial in k .*

(In probabilistic terms this means that $\{S_x : x \in \mathbb{R}^m\}$ is a VC-class or Vapnik-Chervonenkis class; when $\mathbb{R}^n$ is equipped with a probability measure it implies that the law of large numbers holds uniformly for all sets S_x , with bounds that are independent of x .)

All the above actually goes through when instead of the real line $\mathbb{R}$ we take any dense linearly ordered set without endpoints. The notion of o-minimal structure on such an ordered set is defined in the same way as for $\mathbb{R}$. (A basis for the topology on the ordered set is given by the intervals (a, b) , with the product topology on cartesian powers; the usual notion of “connected” is replaced by “definably connected”, which makes no difference when the underlying ordered set is $\mathbb{R}$.) This kind of generalization may seem at first unmotivated, but is actually crucial to prove certain results in the classical real case. We discuss this later in the introduction.

The case that $\mathcal{S}$ contains addition and multiplication (Chapters 6–10)

Up till now we have considered a completely arbitrary o-minimal structure $\mathcal{S}$ on $\mathbb{R}$, and we did not even assume that $\mathcal{S}$ contains (the graph of) addition. *Assume $\mathcal{S}$ contains addition*. Then we have a very useful definable choice principle based on the possibility of singling out the midpoint of a bounded interval. Special cases of this general principle are the “curve selection lemma”, and the fact that a definable equivalence relation on a definable set has a definable set of representatives. These facts have numerous consequences (see Chapter 6) but for deeper results we need also the presence of multiplication. *Assume $\mathcal{S}$ also contains multiplication*.

Then in the cell decomposition theorem we can require the cells to be C^1 -manifolds, and even C^2 -manifolds, and so on (see Chapter 7). (However, for all I know it may be that finer and finer partitions are required as the desired degree of smoothness is increased.) We use C^1 -cell decomposition to prove a “good directions lemma”:

Given a definable set $A \subseteq \mathbb{R}^{n+1}$ of dimension $< n + 1$ there is a unit vector $u \in S^n$ such that every affine line in $\mathbb{R}^{n+1}$ with direction u intersects A in only finitely many points.

This is a substitute for Noether normalization in the proof of the following fundamental result in Chapter 8.

TRIANGULATION THEOREM. *Each definable set is definably homeomorphic to a bounded semilinear set.*

One curious consequence of the triangulation theorem is that two definable sets are definably equivalent (that is, there is a definable, not necessarily continuous, bijection between them) if and only if they have the same dimension and the same Euler characteristic. There are numerous other consequences, such as a definable version of the Tietze extension lemma (Chapter 8, Section 3).

Call a definable map $f: E \rightarrow B$ between definable sets **definably trivial** if there are a definable set F and a definable homeomorphism $E \rightarrow B \times F$ such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{\sim} & B \times F \\ & \searrow f & \swarrow \\ & B & \end{array}$$

commutes. (Note that then all fibers $f^{-1}(b)$ are definably homeomorphic to F .) In Chapter 9 we prove

TRIVIALIZATION THEOREM. *Let $f: E \rightarrow B$ be a definable continuous map between definable sets E and B . Then B can be partitioned into definable sets $B_1, \dots, B_k$ such that the restrictions*

$$f|_{f^{-1}(B_i)}: f^{-1}(B_i) \rightarrow B_i, \quad i = 1, \dots, k,$$

are definably trivial.

As a corollary, a definable collection $\{S_x : x \in \mathbb{R}^m\}$ of definable sets contains only finitely many definable homeomorphism types. Another standard consequence is the conical structure of a definable set in some ball around any of its points.

In combination with the o-minimality of the exponential field of real numbers (Wilkie's theorem) a variant of the trivialization theorem leads to the proof of a conjecture by Benedetti and Risler (Chapter 9, Section 3).

In the final chapter, Chapter 10, we show that under rather weak assumptions definable sets can be glued together by means of definable continuous transition maps. We also use the triangulation theorem to form quotient spaces in the category of definable sets and definable continuous maps.

The triangulation and trivialization theorems go through for each real closed field equipped with an o-minimal structure that contains the addition and multiplication of the field. In fact we prove these results directly in this setting, adapting the treatment of the semialgebraic case by Delfs and Knebusch [11]. This kind of generalization is done not for its own sake but to throw light on the classical real case: certain real closed extensions of $\mathbb{R}$, like the field of algebraic Puiseux series

over $\mathbb{R}$, appear naturally as real closures of function fields of real algebraic varieties. More generally, if $\mathbb{R}$ and a real closed extension R are equipped with corresponding o-minimal structures, then points in R^n serve as generic points of definable subsets of $\mathbb{R}^n$. (This idea will only play a minor role here, but will become quite useful in a later volume where the model-theoretic tools to handle such situations are made available.) One could also argue that some real closed fields, like the field of real algebraic numbers and, at the other extreme, Conway's field of surreal numbers, are of independent interest. (In contrast to real closed fields in general, Conway's field perhaps supports a *canonical* exponential function for which Wilkie's theorem on exponentiation goes through.)

This finishes our preview of the main results. We actually prove stronger and more detailed versions of these theorems, and we also include exercises developing some material a little further. To read this book only a rudimentary knowledge of real analysis and point set topology is needed. In particular, no model theory is used, although the subject of o-minimal structures developed in close contact with model theory. The deeper results that will be treated in a later volume do require some elementary results from model theory, valuation theory, and differential topology.

Notes, comments, and references

Artin and Schreier [1], Koopman and Brown [36] and Tarski [61] are at the origin of much of our subject. Standard references for real algebraic and semialgebraic geometry are the books by Brumfiel [7], Bochnak, Coste and Roy [4], and Benedetti and Risler [2]; see also the articles by Bochnak and Efroymsen [5], Delfs and Knebusch [11], and Knebusch [34]. The theory of semianalytic and subanalytic sets was created by Łojasiewicz [40], Gabrielov [26] and Hironaka [30]; see also Bierstone and Milman [3], Denef and Van den Dries [15], and Sussmann [60], for expositions and alternative treatments. Grothendieck's proposed *topologie modérée* in [28] would contain all those theories as special cases and seems very much in the spirit of o-minimality. The appendix in MacPherson [42] also puts forward a general view on tame topology. It was observed in [20] that the theory of subanalytic sets falls under the scope of o-minimality; see also Van den Dries and Miller [23] for more on this, and for an exposition of o-minimality in the more geometric language of manifolds.

Arbitrary o-minimal structures on the real line were first considered in [19], where they were called "structures of finite type". The term "o-minimal structure" was suggested by Pillay and Steinhorn [49], because of an analogy with the notion of "strongly minimal structure" from model theory. This analogy has since turned out to be very suggestive though it plays no role in this book. The article [49] and its sequels [35] and [48] systematically develop the theory of o-minimal structures on arbitrary linearly ordered sets from a model-theoretic viewpoint. This direction found a recent culmination in Peterzil and Starchenko [46]. See also the survey paper [21] for a description of the various roles of model theory in the study of o-minimal structures. Chang & Keisler [9] is a basic reference for model theory.

An important event was Wilkie's proof in 1991 that the real exponential field is model-complete and o-minimal, see [64]. This proof cleverly used model-theoretic properties of o-minimal structures, and in return gave new insight into how to establish model-completeness and o-minimality of certain structures on the real field. A theorem of Khovanskii [32] on zerosets of exponential polynomials also plays a key role in [64]. Khovanskii's book [33] elaborates on his theorem and contains interesting applications and open problems.

For other o-minimal structures on the real field, see Van den Dries, Macintyre and Marker [22] and Miller [44]. Building o-minimal structures on the real field has become recently (1996-1997) a very lively activity: among the kinds of functions that turn out to generate o-minimal structures are functions defined by Dirichlet series and by multisummable power series, and Pfaffian functions. (I omit references since at the time of writing much of this is only available in the form of preprints and the dust has not settled yet.)

The following information came too late to include in the references at the end of the book:

1. "Esquisse d'un Programme" has finally appeared (with English translation) in

Geometric Galois Actions. 1. Around Grothendieck's Esquisse d'un Programme, edited by L. Schneps and P. Lochak, Cambridge University Press, 1997. (Teissier's article there discusses tame topology.)

2. Tame topology (in the real setting) is also developed extensively in the book

M. Shiota, *Geometry of subanalytic and semialgebraic sets*, Birkhäuser, Boston, Mass., 1997.

